Applied AI Engineer: Home Assignment

The situation

An online retailer processes a few hundred refund requests a week. A crude auto-approve rule handles some of the small ones. Nobody quite remembers its logic. A person reads everything else. Most of those don't need to be read: someone returns a $20 shirt in the wrong size and a human spends two minutes clicking approve. Meanwhile the requests that genuinely need judgment sit in the same queue as everything else.

They want an AI agent to handle the easy ones. The hard part isn't the agent.

What you're starting with

Four CSV files, in the same folder as this brief: data folder

- **history.csv**: about three months of decided requests. For each one you get what was decided, by whom, and what happened afterwards.
- **pending.csv**: requests waiting for a decision.
- **data_dictionary.csv**: what each column is supposed to contain.
- **policy_notes.csv**: the Refund Operations Manager's policy, in her own words.

The files are a sample, not the full volume. Some of the data is messy, and a few of the decisions were wrong.

Part 1 · Decide what the system handles

Work out where the line sits between what gets approved automatically and what goes to a person. Then build the system that applies it.

The system itself can be thin. We're more interested in how you arrived at the line, and how you'd know if it were wrong, both before you switch it on and after. Assume every approval moves real money.

Part 2 · Build the two mechanisms

This is the real assignment.

For the reviewer, the person who picks up the requests the system didn't take. Where does the work land? What do they see? What are their options, and what happens after they decide?

For the Refund Operations Manager, the person who owns the policy. She needs to change what the system does without asking an engineer. For example: *"From now on, stolen-card cases go straight through." "From now on, anything over $300 comes to us."* Expect others. How does she make a change, and how does she know what it will do before it goes live?

AI in your solution

This is an applied AI role, so we want to see your judgment about AI, not just your use of it. Use an LLM where it earns its place, and tell us where you deliberately didn't.

Scope

Aim for 4–6 hours. You won't finish everything, and that's expected. A narrow thing that runs, plus a clear list of what you cut, beats a wide thing that half-works.
Don't build auth, cloud deployment, or a real payments integration.

What to send

A link to a repo with the items below. The write-up and the screen recording can sit in the repo or be linked from the README.

- **The code.** We should be able to run it ourselves with one command, on `pending.csv` or on a new file with the same columns. Include tests.

- **A demo for business users of the two mechanisms.** Something we can click through:

    - The reviewer's queue: what lands there, what they see, what they can do.
    - The manager changing a rule and seeing what it will do before it goes live.

    A simple local web app, Streamlit or a hosted link is fine. If any part is a mock rather than wired to your system, say which. Add a 3–5 minute screen recording, in case we can't run it.

- **A README.** How to run it (including how to plug in an API key), a short architecture sketch, and a line on how you used AI coding tools, including one thing they produced that you changed or threw away.

- **A short write-up,** Max 8 slides. Cover:

    - **The process today:** who's involved, where time and money go, and where it breaks. A rough before/after map is enough.
    - **The line:** what the system handles alone, what goes to a person, and the evidence for it.

- **How you'd measure success:** the few numbers you'd watch, what "good" looks like, and what would make you roll back.
- **Where AI is and isn't used, and why:** the model you chose, a rough cost per decision, and your evaluation results.
- **What you deferred, and what you'd do next.**

Send it to Yaniv Tross (yaniv.tross@wnwd.com) and Gal Ohayon (gal.ohayon@wnwd.com) within 2 days.